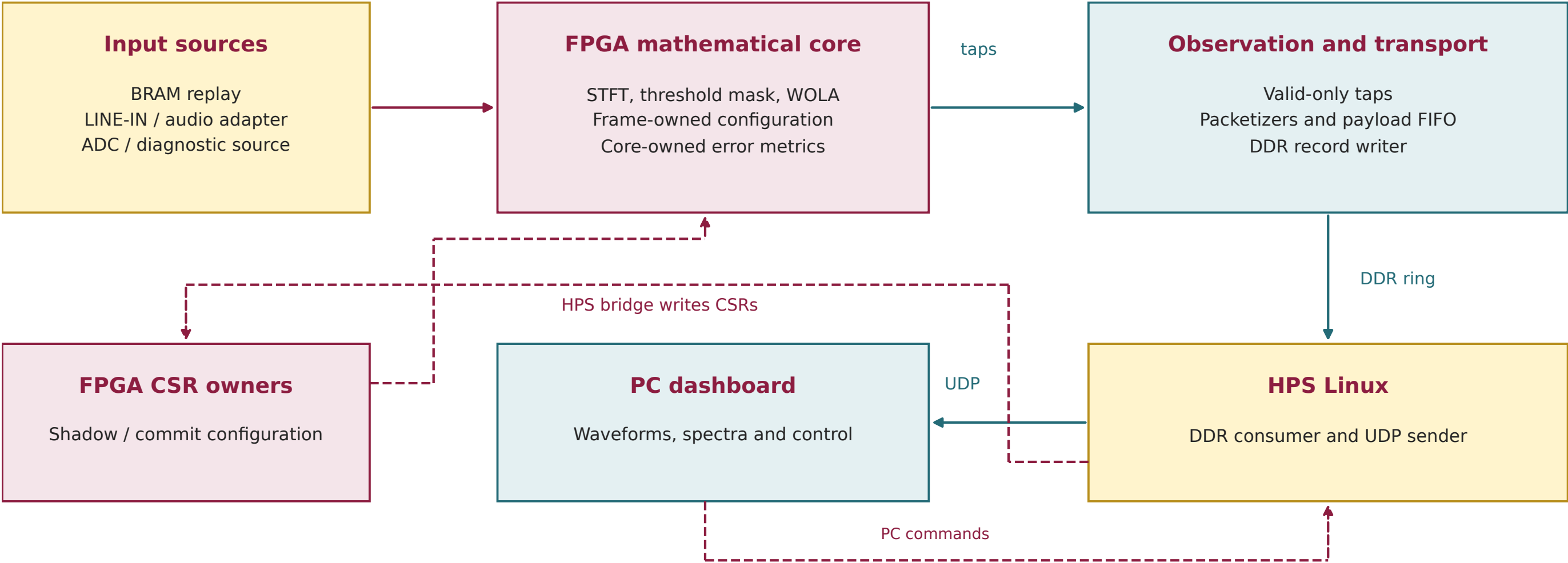


System data flow and ownership

One mathematical core in FPGA fabric, with separate observation and control paths

FPGA core and fabric control: 50 MHz

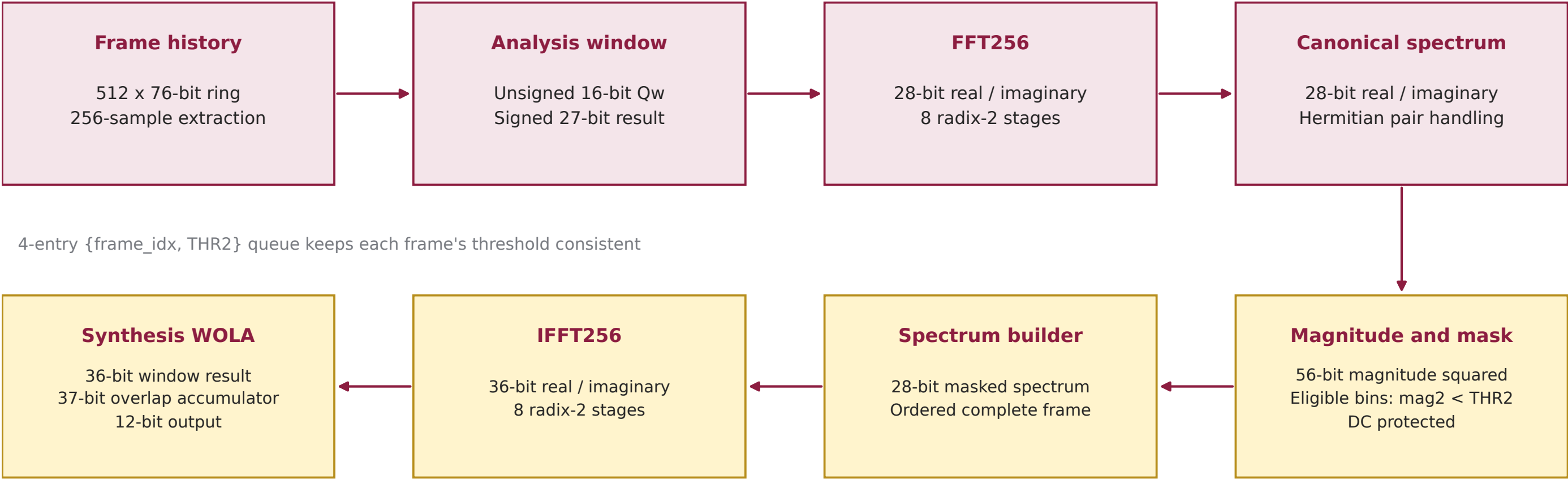


Telemetry congestion may drop records. It cannot backpressure the mathematical core.

Fixed-point DSP datapath

FFT length 256, hop 128, reference alignment D = 384 samples

Signed 12-bit input samples

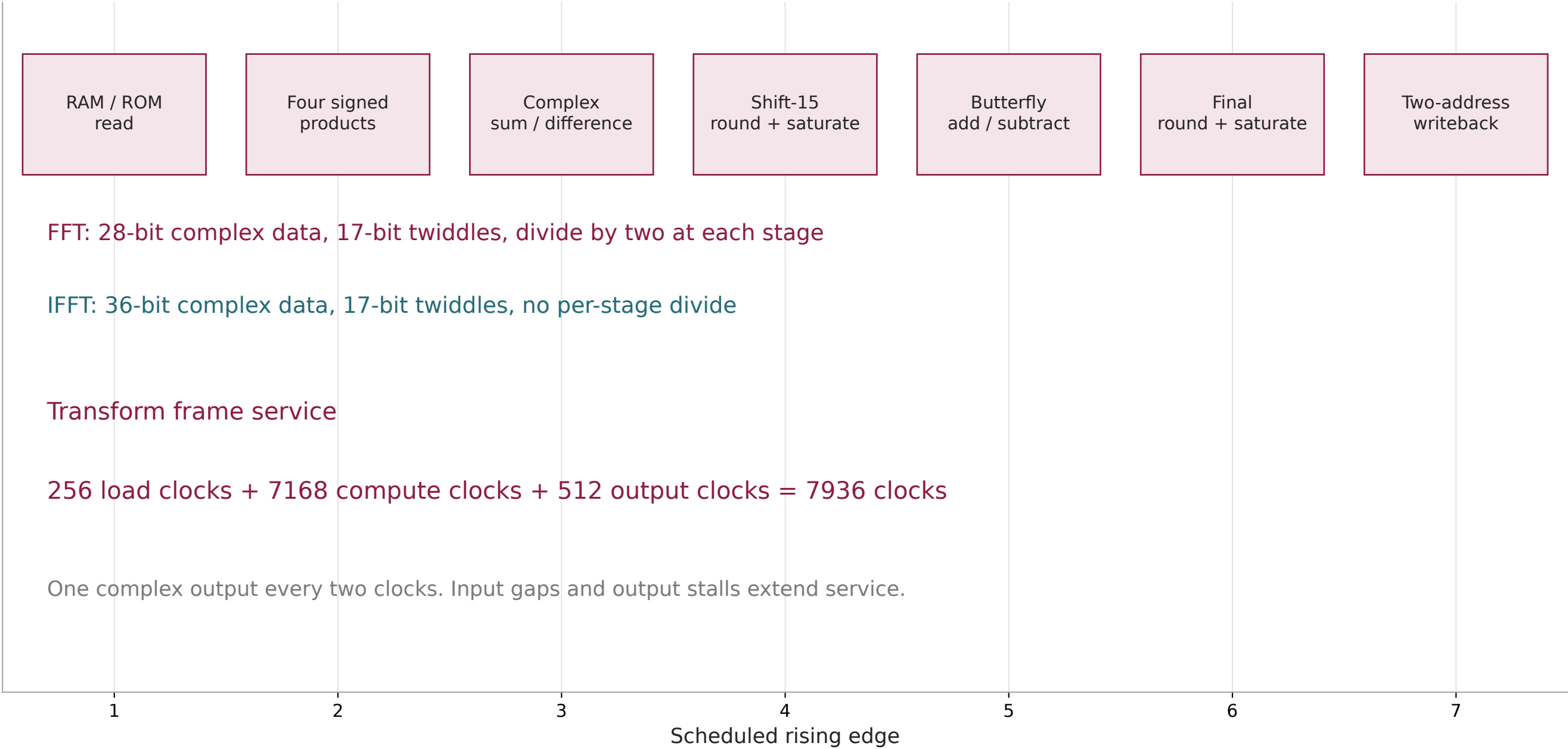


4-entry {frame_idx, THR2} queue keeps each frame's threshold consistent

Delayed input history: 1024 x 76 bits $e[n] = x[n-384] - y[n]$ Full-stream error aggregates

Registered butterfly schedule

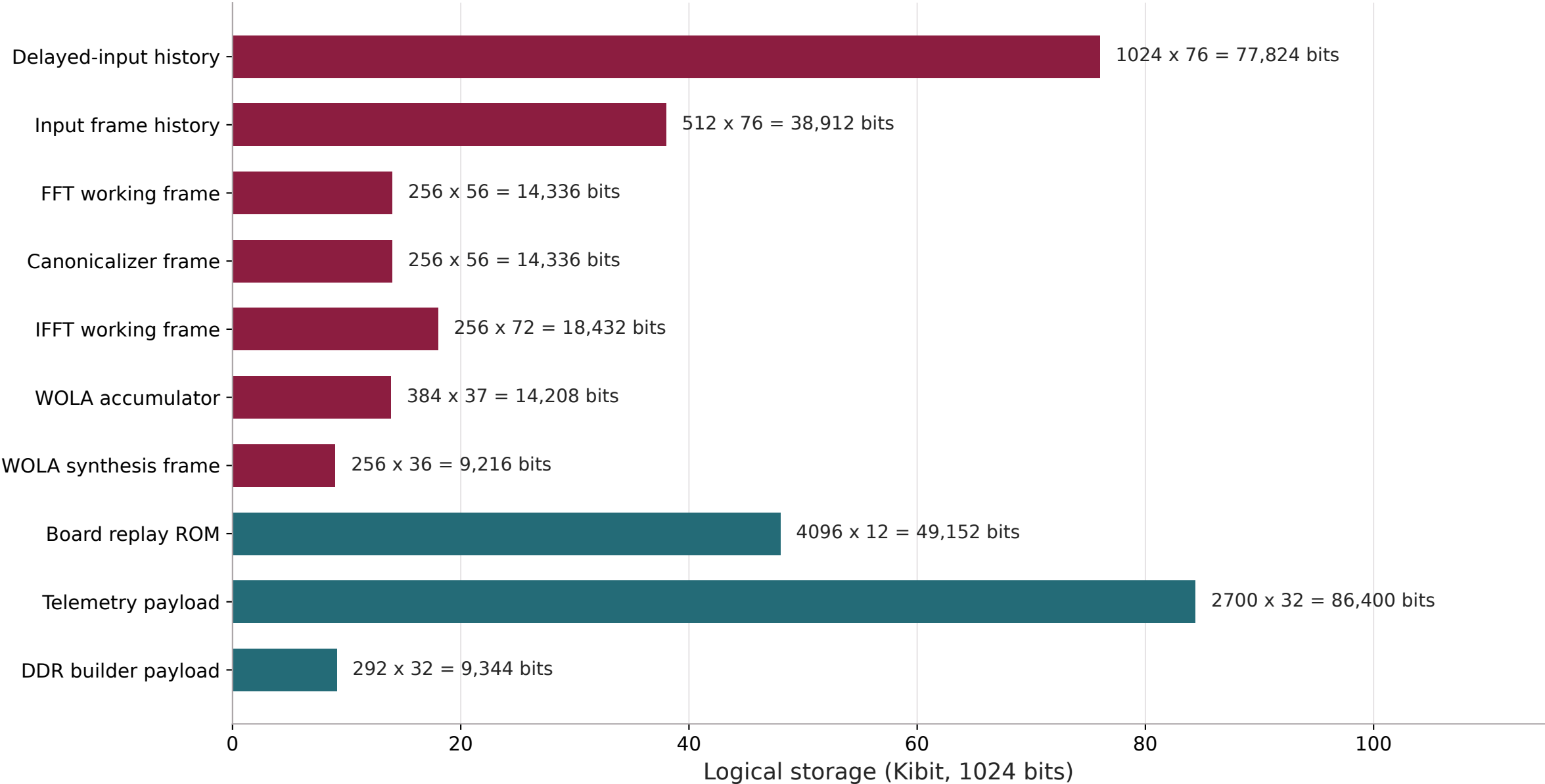
One butterfly uses seven rising edges. Each transform executes all 1024 butterflies per frame.



Source-derived local schedule. Assumes a legal frame, continuous input and ready output.

Selected storage in the implemented architecture

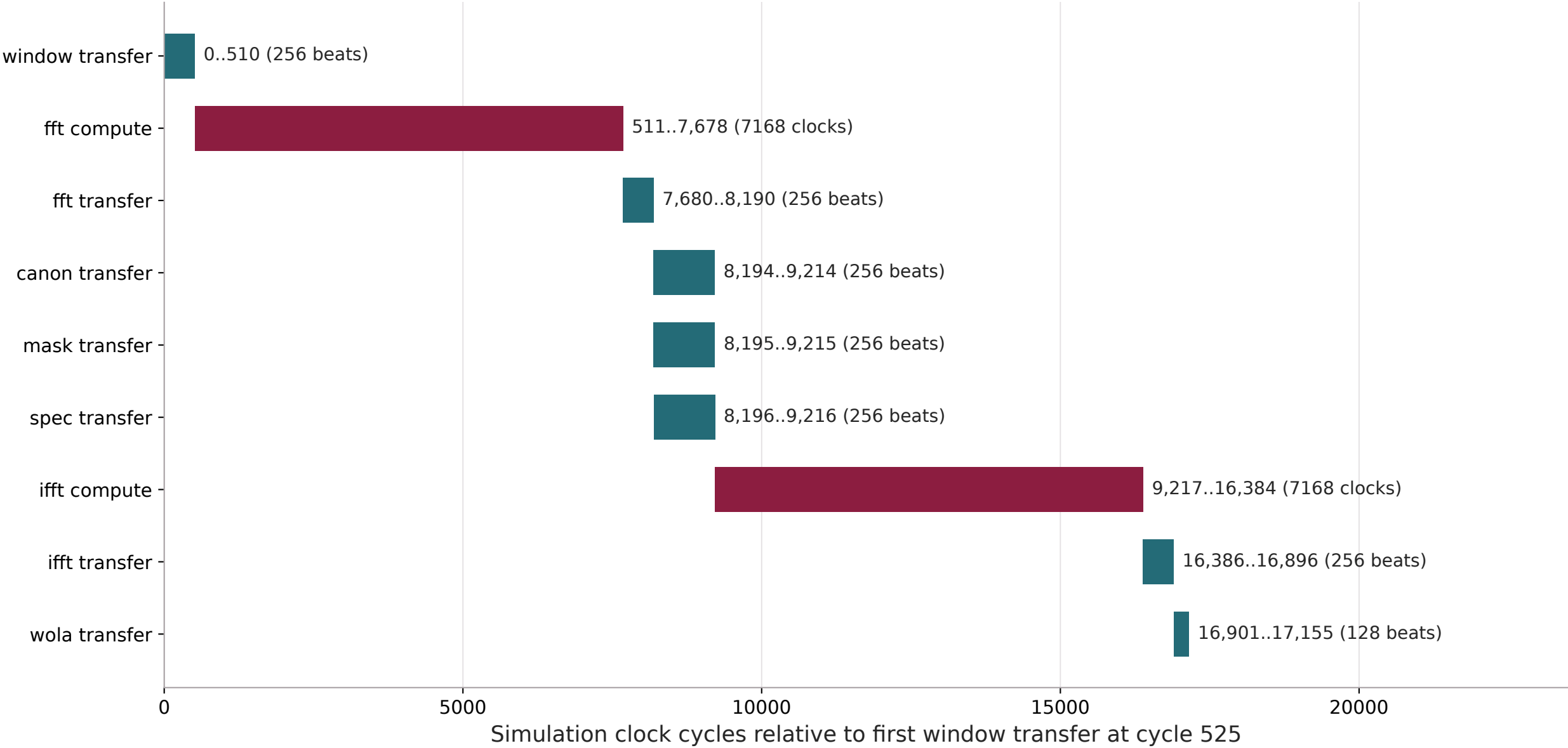
Logical array capacities from RTL. Coefficient ROMs, audio FIFOs and control state are omitted.



Logical capacities do not predict fitted M10K packing. The complete fitted board uses 55 RAM blocks.

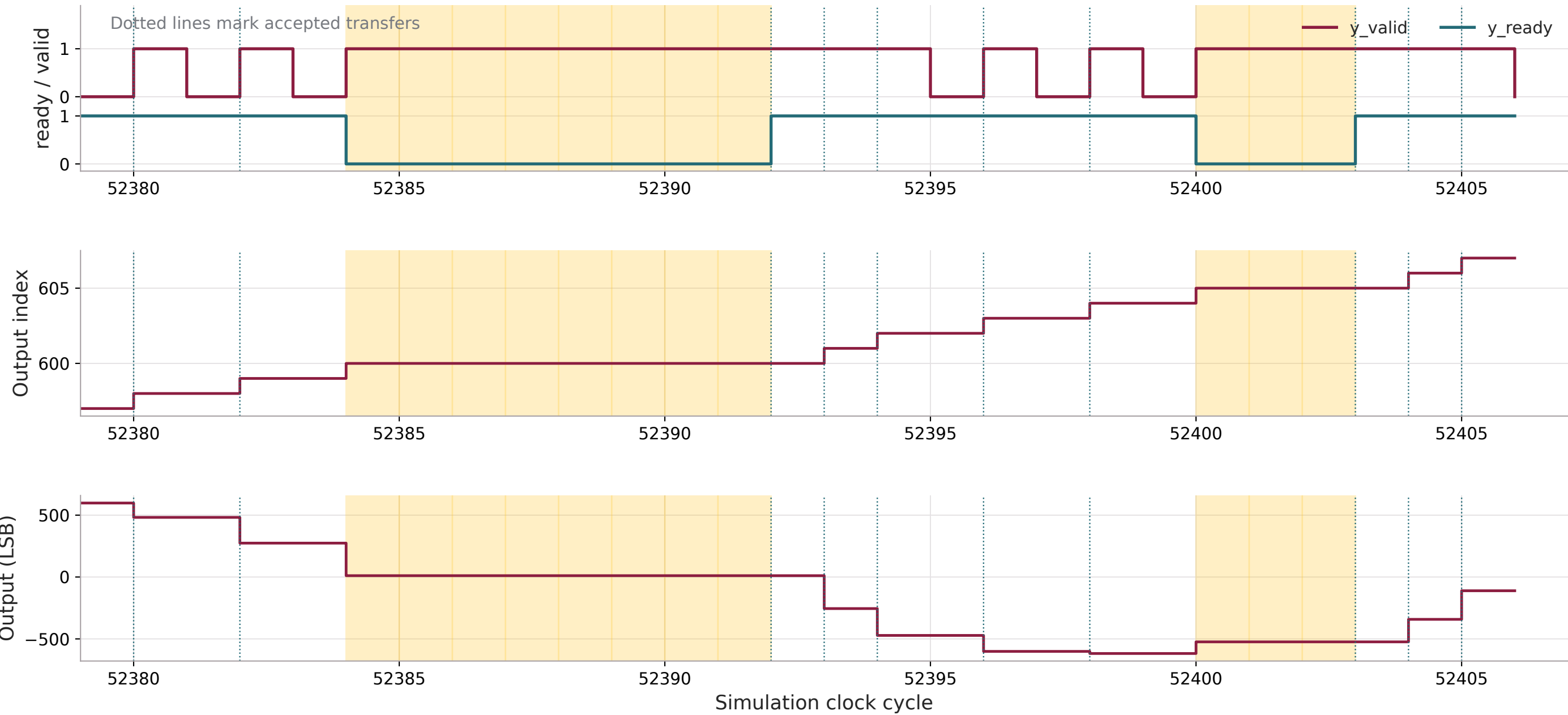
Observed first-frame pipeline activity

Fresh ModelSim core simulation. Interface spans and transform compute phases come from recorded RTL signals.



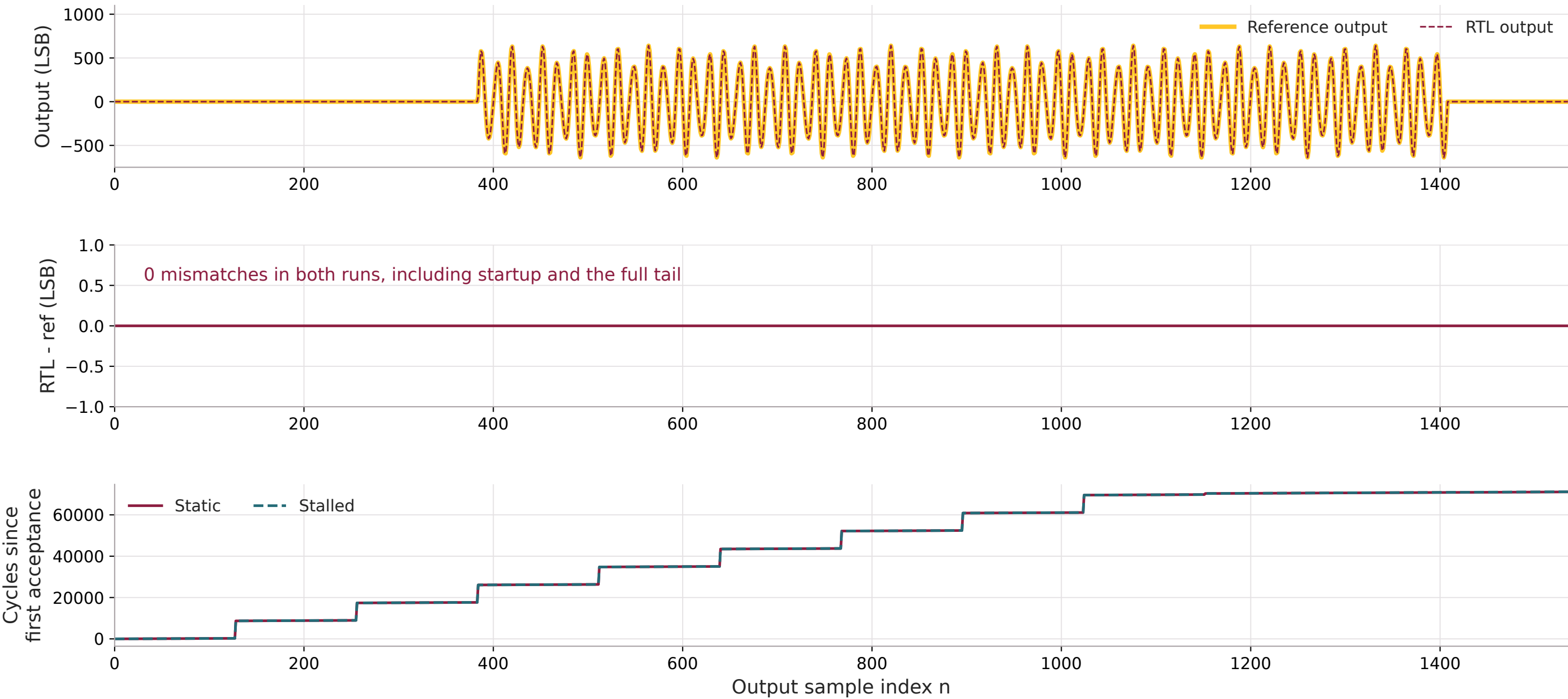
Output handshake under backpressure

Actual signals from the stalled RTL run. Transfer occurs only when valid and ready are both high.



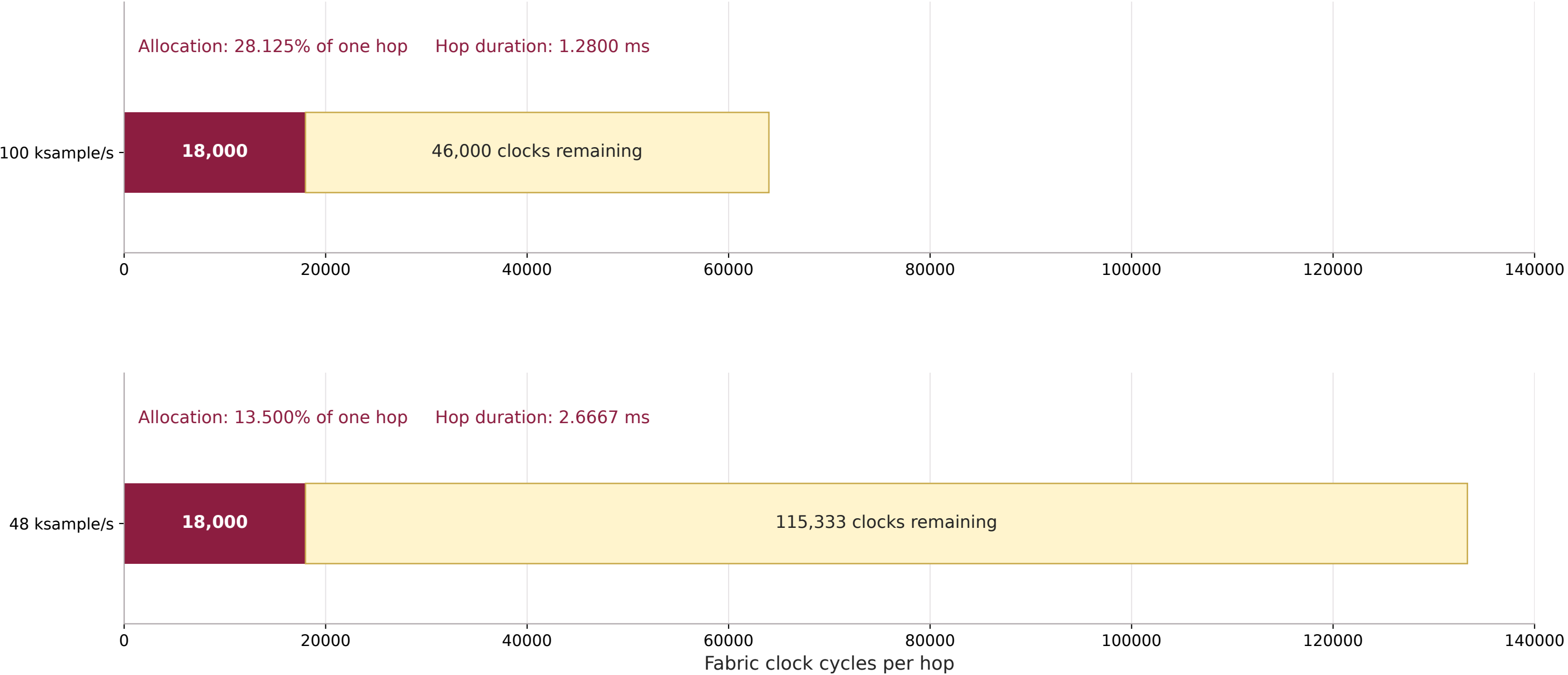
RTL and reference output waveforms

Multitone input, THR2 = 4096, 1536 output samples. Static and stalled runs preserve the same output codes.



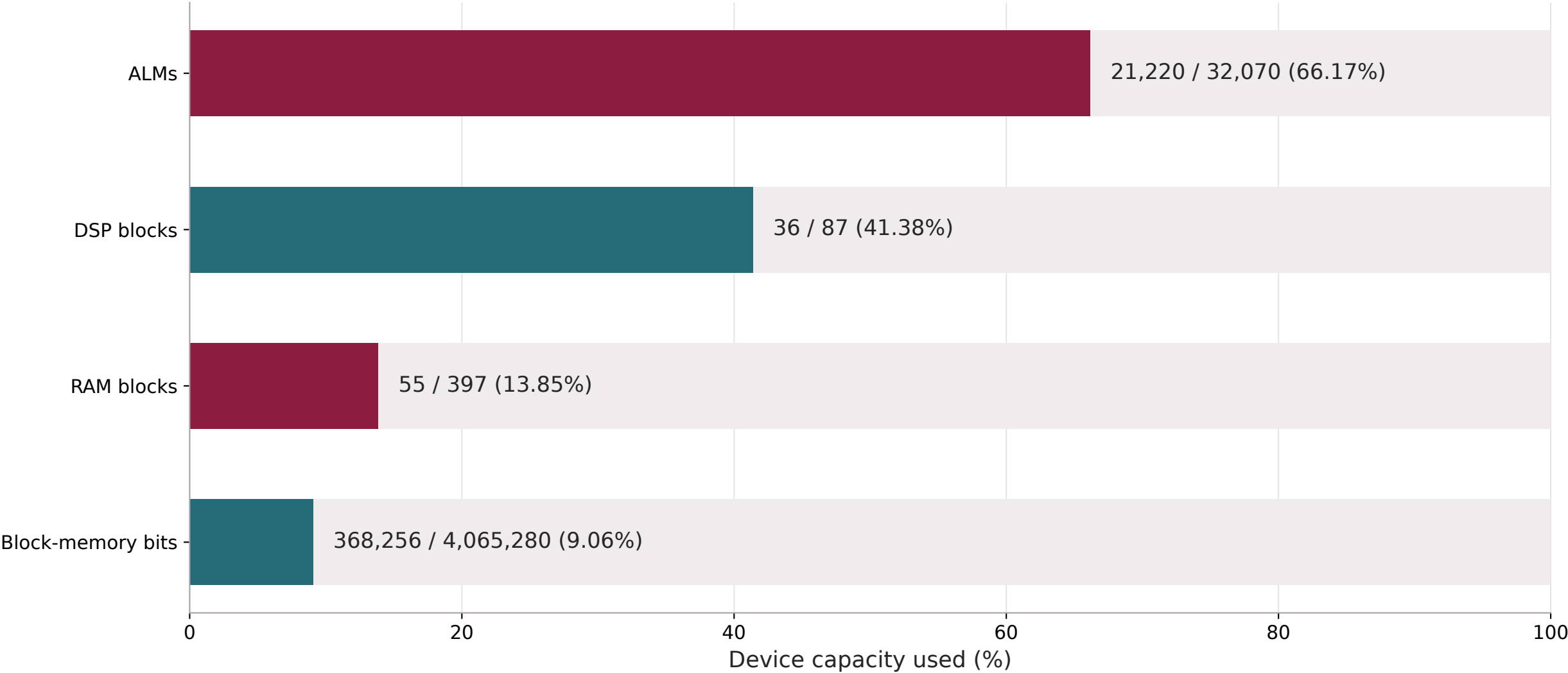
Frame-work allocation and sample-rate deadlines

50 MHz fabric, 128 samples per hop. The 18000-clock allocation covers the documented connected schedule.



Fitted FPGA resource utilization

Quartus Prime Lite 20.1.1, Cyclone V 5CSEMA5F31C6. Archived BRAM build dated 15 September 2026.



29,452 registers 209 fitted pins checked Whole-board BRAM profile

Fitted timing across four operating corners

CLOCK_50 to CLOCK_50 at a 20 ns clock period. All four corners use the 1100 mV model.

